

OMAR ASHRAF

Senior AI/ML Engineer

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PROFILE

Senior AI/ML Engineer with 6+ years of experience building and deploying production-grade machine learning systems and LLM-powered applications. Deep expertise in multi-agent architectures, RAG pipelines, voice AI, computer vision, and full-stack AI integration across legal SaaS, EdTech, healthcare, e-commerce, and document/identity automation. Track record of taking systems from prototype to production with measurable outcomes.

WORK EXPERIENCE

ML Engineer | Turing, USA Oct 2024 – Present | Remote, Part-Time

- Built and evaluated LLM training datasets across 10+ projects spanning code generation, algorithmic problem-solving, function-calling review, and physics simulation, covering graph algorithms, FFT-based multiplication, quantum walk simulators, and 2-SAT constraint solving.
- Developed a full-stack RL Gym Harness (FastAPI + Celery + PostgreSQL + React) to benchmark Computer Using Agent models from OpenAI, Anthropic, and Gemini on real-world UI automation tasks, including an AI-powered insights layer that auto-generated per-execution and per-batch failure summaries with leaderboards and exportable Excel reports.
- Authored comprehensive unittest suites for complex algorithmic systems (energy distribution analysis, polynomial rolling hash, Steiner tree optimization), covering every edge case, constraint, and formula from the prompt spec with no mocking or decomposed assertions.
- Curated LiveCodeBench and BigCodeBench problem sets: designed hard coding prompts that broke nine AI models, wrote ideal solutions with 14+ test cases at 90%+ branch coverage, and standardized public/private test case formatting across the benchmark pipeline.
- Prepared code interpreter LLM training data, writing natural language prompts, producing step-by-step verified solutions, executing code in terminal environments, and submitting qualitative feedback on model outputs across GitHub repository modification and RLHF annotation tasks.

Senior AI/ML Engineer | Solgenci 2023 – Present

- Built CasePeer, a legal SaaS platform, using a 5-agent LangGraph architecture with Gemini OCR, n8n OTP workflows, PostgreSQL checkpointing, and a REST API, automating legal document processing end-to-end and cutting case processing time by over 60%.
- Engineered a Real Estate SMS Lead Qualification pipeline (Django 5, DRF, Celery, Redis, SignalWire, Gemini 2.5 Flash, Follow Up Boss CRM) handling 500+ leads/day with sub-3-second response times, and a Planet Beauty Shopify/ShipStation CRM integration with Azure OAuth2 and n8n automation that eliminated around 70% of manual CS work.
- Set up on-prem GPU infrastructure for clients with data sovereignty requirements: RTX 4090 servers on Ubuntu with Nginx reverse proxy and SSH tunneling, deploying vLLM and Ollama to serve Llama 3, Mistral, and Qwen with AWQ/GGUF/GPTQ quantization and multi-GPU tensor parallelism.
- Developed an AI Structured Learning Platform with GPT-4 + Pinecone RAG, Pydantic validation, asyncio concurrency, Supabase, AWS S3, Docker, and GitHub Actions CI/CD, indexing 1,000+ educational PDFs with metadata filtering by subject and grade.
- Delivered ZÉROLIMITE, a French-language AI coaching diagnostic for a Quebec academy: a 36-node n8n workflow with 6 Claude-powered agents across 5 coaching stages, Supabase state persistence, and PDF report output.
- Implemented GraphQL for optimized data fetching across AI model integrations, reducing data load times by 40%, and streamlined backend systems using FastAPI, improving server response times by 20%.
- Worked directly with US, UK, and Australian client stakeholders: scoped requirements, ran technical walkthroughs, and advised on infrastructure and model selection decisions.

Data Scientist | Senarios 2021 – 2023

- Built Convirza, a HIPAA-compliant AI conversation analytics platform: a hybrid ASR pipeline combining AssemblyAI, Deepgram, and Whisper, BERT fine-tuning and LoRA-adapted LLMs for sentiment and call-outcome scoring, served via TorchServe on GKE. Sentiment model hit ~90% accuracy (3-class) with an F1 of ~0.88 on the minority class, and word error rate on phone audio came in around 8–10%, a ~30% improvement over the legacy transcription system.

- Delivered BundleSkills (EdTech): a 6-agent LangGraph system (Orchestrator, Research, Content, Validation, Enrichment, Diff) on Celery + Redis with 4 separate Pinecone indexes (curriculum, source, taxonomy, reference) and Pydantic v2 schema validation as a hard gate, reducing skill-unit production time from 3-5 days to under 20 minutes with a 90%+ first-pass validation rate.
- Built intelligent chatbots using OpenAI and LangChain for multiple clients, including a tax advisory chatbot with deep learning-based intent classification, and an end-to-end MCQ and content generation pipeline for a kids EdTech platform using GPT fine-tuning tailored to subject, topic, and student profile.
- Fine-tuned Stable Diffusion 2.1 using Dreambooth and Textual Inversion for a text-to-image product (Joxy.ai) across single and multi-GPU setups, handling dataset curation and prompt monitoring throughout training.
- Built scalable Python AI pipelines reducing resource consumption by 20%, and led a model training pipeline overhaul that improved data handling and processing speed across all active projects.

Junior Data Scientist | Machine Learning 1 2019 – 2021

- Built RecordMe.ai, a generalized invoice, receipt, and bookkeeping automation extractor: trained and deployed LayoutLM, TrOCR, and Donut document-understanding transformers for automated field extraction, with a custom BERT-based NER layer for address parsing, deployed via FastAPI on a containerized Ubuntu server.
- Built the ID card segmentation module of an end-to-end KYC pipeline using U-Net trained on 800+ annotated images, achieving ~0.98 IoU on validation and enabling precise OpenCV perspective correction before downstream processing.
- Developed the field detection layer using YOLOv5, trained on 500+ labeled ID images across field types (Name, DOB, Address, Photo, ID Number), hitting ~95% mAP and feeding OCR with semantically labeled crops.
- Built the fraud detection layer: a Mask R-CNN model to segment the face photo region followed by a ResNet-18 binary classifier for photo replacement detection, plus a separate 3-class presentation attack detector (ResNet-50) distinguishing real IDs from printed copies and screen replays.
- Integrated Google Cloud Vision OCR for field text extraction with spaCy NER and regex validation for name, date, and document number fields, overlapping async OCR calls with CNN inference via FastAPI + asyncio to keep total pipeline latency under 1 second on a V100.
- Deployed the full KYC pipeline to production: containerized all models in Docker (PyTorch and TensorFlow in the same image on a CUDA runtime base), set up FastAPI behind Nginx, and managed GPU memory allocation to avoid framework conflicts at runtime.
- Built edge AI pipelines in TensorFlow/Keras for Android and iOS, achieving ~98% face recognition accuracy with anti-spoofing and ambient light liveness detection, and improved search and recommendation accuracy by 15% through custom ranking filters and embedding-based retrieval.

SKILLS

LLM Frameworks: LangGraph, LangChain, LlamaIndex, OpenAI, Anthropic Claude, Gemini, Mistral, Llama-2

ML/DL: PyTorch, TensorFlow, Keras, Hugging Face Transformers, BERT fine-tuning, LoRA, LayoutLM, TrOCR, Donut, U-Net, Mask R-CNN, YOLOv5, CNN, RNN, LSTM, Stable Diffusion

Backend & APIs: FastAPI, Django, Flask, GraphQL, REST, WebSockets, Celery, Redis, n8n

Data & Vector Stores: PostgreSQL, MongoDB, Neo4j, Pinecone, Supabase, MySQL

Speech/Voice AI: AssemblyAI, Deepgram, Whisper, Twilio, SignalWire

Cloud & DevOps: AWS (EC2, S3, ECS, SageMaker, API Gateway), GKE, Docker, GitHub Actions, CI/CD, Azure

Computer Vision: OCR (Google Vision, Tesseract), NER (spaCy), object detection, background removal, anti-spoofing

EDUCATION

BS Mechatronics Engineering

University of Engineering and Technology, Lahore